

1 Question (10 points)

Given the following function

```
int algorithm1(int N){
    int sum = 0;
    for (int i = 1; i <= N; i++){
        for (int j = i; j <= N; j++){
            for (int k = j; k <= N; k++){
                sum++;
            }
        }
    }
}
```

What is the value of sum after the execution of the algorithm1?

2 Question (12 points)

You are given an integer matrix of $A[][]$, which is the adjacency matrix of a graph. Write a function that checks if the graph has the circular topology.

```
boolean circularTopology(int[] [] A)
```

3 Question (14 points)

Write a recursive pseudocode for computing the length of the longest path to a leaf node in a binary tree.

4 Question (14 points)

Solve the following version of the knapsack problem. Given n items of known weights $w_1, \dots, w_n$ and values $v_1, \dots, v_n$ and a knapsack of capacity W , find the most valuable subset of the items that fit into the knapsack. Assume that there are unlimited quantities of copies for each of the n item kinds given.